

CSA0614-TOPIC 5-DAA

Y Greeshma Reddy

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SIMATS ENGINEERING SIMATS Online Programming Lab DAA PROGRAMMING YANAMALA GREESHMA REDDY

LAB PROGRAM- TOPIC 5

Language Python C++ Java

QUESTIONS

Q1 Search an Element in a Sorted Array 10 marks

Q2 Find the Index of a Number in Binary Search 10 marks

Q3 Count the Number of Occurrences of a Number in Binary Search 10 marks

Q4 Find the First Occurrence of a Number 10 marks

Q5 Find the Last Occurrence of a Number 10 marks

Q6 Find and Count Repetitive Numbers 10 marks

Q7 Find the Index of a Number in Binary Search 10 marks

Q8 Find the First Occurrence of a Number 10 marks

Q9 Find the Last Occurrence of a Number 10 marks

Q10 Find and Count Repetitive Numbers 10 marks

10/10 answered

Search an Element in a Sorted Array 10 marks

Given a sorted array of integers and a key value, use the Binary Search (Divide and Conquer) method to determine whether the key is present in the array. If found, print its position otherwise, print "Element not found".

Your Code

```
def binary_search(arr, key):
    low = 0
    high = len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1

# Example usage
arr = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19]
key = 5
index = binary_search(arr, key)
if index != -1:
    print(f"Element found at position {index}")
else:
    print("Element not found")
```

Input

2 4 6 8

Output

Element found at position 2

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Q8 Find the First Occurrence of a Number 10 marks

Q9 Find the Last Occurrence of a Number 10 marks

Q10 Find and Count Repetitive Numbers 10 marks

10/10 answered

Find the Index of a Given Number 10 marks

Given a sorted array and a target number, find the index of the target using Binary Search. Assume array indexing starts from 0.

Your Code

```
def binary_search(arr, target):
    low = 0
    high = len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return -1

# Example usage
arr = [1, 3, 5, 7, 9, 11, 13, 15, 17, 19]
target = 5
index = binary_search(arr, target)
print(f"Element found at index {index}")
```

Input

5

2 4 6 8 10

Output

Element found at index 2

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LAB PROGRAM - TOPIC 5

Language: Python C++ Java

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Q5 Find the Last Occurrence of an Element 10 marks

Q6 Find the First and Last Occurrence of an Element 10 marks

Q7 Find the First and Last Occurrence of an Element 10 marks

Q8 Find the First and Last Occurrence of an Element 10 marks

Q9 Find the First and Last Occurrence of an Element 10 marks

Q10 Find the First and Last Occurrence of an Element 10 marks

10/10 answered

Count the Number of Iterations in Binary Search 10 marks

Given a sorted array and a key, perform Binary Search and display the number of iterations required to find the element.

Your Code

```
def binary_search_with_iterations(arr, key):
    low = 0
    high = len(arr) - 1
    iterations = 0

    while low <= high:
        iterations += 1
        mid = (low + high) // 2

        if arr[mid] == key:
            return iterations
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return iterations  # Return total iterations when if element not found

# Taking input
arr = []
# Input Element, space separated array elements
arr = list(map(int, input().split()))
# Input Element, the key to search
key = int(input())

result = binary_search_with_iterations(arr, key)
print(result)

# Example: [1,2,3,4,5,6,7,8,9,10]
# Input: 5
# Output: 4
```

Input

1 2 3 4 5 6 7 8 9 10

5

Output

4

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LAB PROGRAM - TOPIC 5

Language: Python C++ Java

QUESTIONS

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Q5 Find the Last Occurrence of an Element 10 marks

Q6 Find the First and Last Occurrence of an Element 10 marks

Q7 Find the First and Last Occurrence of an Element 10 marks

Q8 Find the First and Last Occurrence of an Element 10 marks

Q9 Find the First and Last Occurrence of an Element 10 marks

Q10 Find the First and Last Occurrence of an Element 10 marks

10/10 answered

Find the First Occurrence of an Element 10 marks

Given a sorted array containing duplicate elements, find the first occurrence of a given key using Binary Search.

Your Code

```
def find_first_occurrence(arr, key):
    low = 0
    high = len(arr) - 1
    result = -1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            result = mid
            # Continue searching to the left to find the first occurrence
            high = mid - 1
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return result

# Example input
arr = [1, 2, 2, 3, 4, 5, 6]
key = 2

result = find_first_occurrence(arr, key)
print(result)

# Output: 1
```

Input

1 2 3 4 5 6

2

Output

1

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LAB PROGRAM - TOPIC 5

Language: Python C++ Java

QUESTIONS

- Q1: Search an Element in a Sorted Array. 10 marks
- Q2: Find the Index of a Binary Number. 10 marks
- Q3: Count the Number of Elements in a Binary Array. 10 marks
- Q4: Find the First Occurrence of an Element. 10 marks
- Q5: Find the Last Occurrence of an Element. 10 marks
- Q6: Highest and Lowest Temperature Recorded. 10 marks
- Q7: Find and Remove Duplicate Elements. 10 marks
- Q8: Merge Two Sorted Arrays. 10 marks
- Q9: Merge and Sort an Array. 10 marks
- Q10: Merge and Sort an Array. 10 marks

10/10 answered

Find the Last Occurrence of an Element 10 marks

Given a sorted array containing duplicate elements, find the last occurrence of a given key using Binary Search.

Your Code

```
def find_last_occurrence(arr, key):
    low = 0
    high = len(arr) - 1
    result = -1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == key:
            result = mid
            low = mid + 1
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return result

# Example usage
arr = [1, 2, 3, 4, 5, 5, 5, 6]
key = 5
print(find_last_occurrence(arr, key))
```

Input: 5 2 3 4 5 5 5 6

Output: 6

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LAB PROGRAM - TOPIC 5

Language: Python C++ Java

QUESTIONS

- Q1: Search an Element in a Sorted Array. 10 marks
- Q2: Find the Index of a Binary Number. 10 marks
- Q3: Count the Number of Elements in a Binary Array. 10 marks
- Q4: Find the First Occurrence of an Element. 10 marks
- Q5: Find the Last Occurrence of an Element. 10 marks
- Q6: Highest and Lowest Temperature Recorded. 10 marks
- Q7: Find and Remove Duplicate Elements. 10 marks
- Q8: Merge Two Sorted Arrays. 10 marks
- Q9: Merge and Sort an Array. 10 marks
- Q10: Merge and Sort an Array. 10 marks

10/10 answered

Highest and Lowest Temperature Recorded 10 marks

A weather station records temperatures over a week. Find the minimum and maximum temperature using the Divide and Conquer approach.

Your Code

```
def find_min_max(arr, low, high):
    # Base case: single element
    if low == high:
        return arr[low], arr[high]

    # Divide the array into two halves
    mid = (low + high) // 2

    # Recursively find min and max for the left and right halves
    left_min, left_max = find_min_max(arr, low, mid)
    right_min, right_max = find_min_max(arr, mid + 1, high)

    # Compare the results
    overall_min = min(left_min, right_min)
    overall_max = max(left_max, right_max)

    return overall_min, overall_max

def main():
    # Taking input from the user
    # Example input: 40 41 32 33
    data = list(map(int, input().split()))

    if len(data) > 0:
        min_temp, max_temp = find_min_max(data, 0, len(data) - 1)

        print(f"Minimum Temperature: {min_temp}")
        print(f"Maximum Temperature: {max_temp}")

    else:
        print("No data available")

if __name__ == "__main__":
    main()
```

Input: 40 41 32 33

Output: Minimum Temperature: 32
Maximum Temperature: 41

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LAB PROGRAM - TOPICS

Language: Python C++ Java

QUESTIONS

- Q1: Search an Element in a Sorted Array (10 marks)
- Q2: Find the Index of a Given Number (10 marks)
- Q3: Count the Number of Occurrences of a Number (10 marks)
- Q4: Find the Most Occurrences of an Element (10 marks)
- Q5: Find the Last Occurrences of an Element (10 marks)
- Q6: Find the Count of Transposition Required (10 marks)
- Q7: Sort and Store Student Marks (10 marks)
- Q8: Stock Price Analysis (10 marks)
- Q9: Sort and Store Stock Price (10 marks)
- Q10: Sort and Store Student Building Heights (10 marks)

10/10 answered

Best and Worst Student Marks (10 marks)

A teacher wants to identify the highest and lowest marks scored in a class using Divide and Conquer.

Your Code:

```
def find_max_min(arr):  
    low, high = 0, len(arr) - 1  
    if low == high:  
        return arr[low], arr[high]  
    mid = (low + high) // 2  
    if arr[mid] > arr[high]:  
        return arr[mid], arr[high]  
    else:  
        return arr[mid], arr[low]  
  
# Recursion Step: Divide the array into two halves  
def div_conq(arr, low, high):  
    max_val, min_val = find_max_min(arr, low, mid)  
    max_val, min_val = find_max_min(arr, mid + 1, high)  
    # Recursion: Compare result of the halves  
    overall_max = max(max_val, max_val)  
    overall_min = min(min_val, min_val)  
    return overall_max, overall_min  
  
def main():  
    n = int(input("Enter the size of array: "))  
    arr = list(map(int, input("Enter the array elements: ").split()))  
    max_val, min_val = find_max_min(arr, 0, n - 1)  
    print(f"Maximum Mark: {max_val}")  
    print(f"Minimum Mark: {min_val}")  
  
if __name__ == "__main__":  
    main()
```

Input: 20 25 27 22

Output: Maximum Mark: 27
Minimum Mark: 20

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LAB PROGRAM - TOPICS

Language: Python C++ Java

QUESTIONS

- Q1: Search an Element in a Sorted Array (10 marks)
- Q2: Find the Index of a Given Number (10 marks)
- Q3: Count the Number of Occurrences of a Number (10 marks)
- Q4: Find the Most Occurrences of an Element (10 marks)
- Q5: Find the Last Occurrences of an Element (10 marks)
- Q6: Find the Count of Transposition Required (10 marks)
- Q7: Sort and Store Student Marks (10 marks)
- Q8: Stock Price Analysis (10 marks)
- Q9: Sort and Store Stock Price (10 marks)
- Q10: Sort and Store Student Building Heights (10 marks)

10/10 answered

Stock Price Analysis (10 marks)

An investor wants to know the minimum and maximum stock prices recorded during a trading week.

Your Code:

```
def main():  
    n = int(input("Enter the size of array: "))  
    arr = list(map(int, input("Enter the array elements: ").split()))  
    min_val, max_val = find_min_max(arr, 0, n - 1)  
    print(f"Minimum Stock Price: {min_val}")  
    print(f"Maximum Stock Price: {max_val}")  
  
def find_min_max(arr, low, high):  
    min_val, max_val = arr[low], arr[high]  
    if low == high:  
        return min_val, max_val  
    mid = (low + high) // 2  
    min_val, max_val = find_min_max(arr, low, mid)  
    min_val, max_val = find_min_max(arr, mid + 1, high)  
    # Recursion: Compare result of the halves  
    overall_min = min(min_val, min_val)  
    overall_max = max(max_val, max_val)  
    return overall_min, overall_max  
  
if __name__ == "__main__":  
    main()
```

Input: 2000 2200 2500

Output: Minimum Stock Price: 2000
Maximum Stock Price: 2500

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DATA PROGRAMMING

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LAB PROGRAM- TOPIC 5

Language Python C++ Java

QUESTIONS

Q1: Search an element in a sorted array. 10 marks

Q2: Find the index of a element. 10 marks

Q3: Count the Number of Occurrences of an Element. 10 marks

Q4: Find the First Occurrence of an Element. 10 marks

Q5: Find the Last Occurrence of an Element. 10 marks

Q6: Find the First and Last Occurrence of an Element. 10 marks

Q7: Find the Most Occurring Element. 10 marks

Q8: Find the Most Frequent Number. 10 marks

Q9: Find the Most Frequent Number. 10 marks

Q10: Find the Most Frequent Number. 10 marks

10/10 answered

Fastest and Slowest Race Times

10 marks

A sports analyst needs to find the fastest (minimum) and slowest (maximum) race completion times.

Your Code

```
def find_fast_slowest_times():
    1. if not times:
    2.     return None, None
    3.     return times[0], times[-1]
    4.     return min(times), max(times)
    5.     return fastest, slowest
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